Methodology — Random Forest ingredient-presence pipeline

Question and scientific boundary

The workflow is a Case Study 1 Data Science project: it predicts the supplied binary `has_risk` alert label from ingredient-list information. It does not infer ingredient toxicity, legal status, product safety, individual allergy risk, concentration, or exposure. The source metadata records only that a 16-item reference count existed; the exact roster, versioned sources, original matching rules, and labeling code are unavailable.

Data acquisition and audit

The supplied snapshots contain 64,237 raw rows, 17,354 interim rows, and 15,621 labeled rows. They are historical artifacts: a French-language export is not evidence of a France-only population. Current API collection is supplementary acquisition evidence, not the origin of the historical labels. Source fields and input data are kept auditable rather than retroactively relabeled.

Cleaning and formula grouping

The workflow safely parses supplied ingredient-list literals, normalizes Unicode/HTML/casing/whitespace, validates binary labels and ingredient counts, and compares parsed lists with raw ingredient text. **47** low-alignment rows are quarantined, leaving **15,574** modeling rows (4,059 positive and 11,515 negative supplied labels). Long or prose-like entries are flagged for review rather than automatically treated as invalid.

A formula group is an order-insensitive fingerprint of normalized unique ingredients. There are 14,832 unique groups and 742 duplicate extra rows. Grouping prevents exact normalized formulas from crossing the internal train/test boundary, but it does not eliminate near-duplicates or demonstrate external generalization.

EDA

EDA inspects missingness, label balance, ingredient-count distributions, categories, country tags, and frequent normalized ingredients. These charts and tables are descriptive and should not be read as toxicological rankings. Category and country metadata can be missing, multi-valued, noisy, or operational rather than legal classifications.

Features and preprocessing

Within each training fold, the pipeline learns the **top 1,500 ingredients** and encodes their presence as binary columns. It adds **log1p(ingredient_count)** . Categories, brands, product names, and registry matches are excluded from the model feature matrix. Feature selection is fit within the relevant training partition to avoid using held-out fold information.

Model comparison, selection, and tuning

Candidate models are compared using **three-fold grouped cross-validation on the training partition**. The comparison evidence is retained transparently. **Random Forest** is the selected ensemble-focused final pipeline and is tuned by **training-CV average precision**. The report field benchmark_cv_leader is shown separately because the overall CV leader can differ from the final Random Forest; the workflow does not retroactively label Random Forest as the winner of every benchmark.

Recorded comparison leader: **Balanced Logistic Regression**.

- Training grouped-CV average precision: **0.9990**.
- Training grouped-CV F1: **99.86%**.

Tuned settings from the completed run:

Parameter	Recorded value
class_weight	balanced_subsample
max_features	0.2
min_samples_leaf	2
n_estimators	200

Threshold selection and final internal evaluation

The final training partition contains 12,459 rows and the held-out partition 3,115 rows, with zero formula-group overlap. Random Forest training out-of-fold probabilities choose the threshold **0.438665** by maximizing **F2** among thresholds that satisfy **recall ⩾ 0.95**. The held-out test does not select the family, parameters, or threshold.

Report both 0.5-default and training-selected threshold results. The primary test is an **internal holdout**, not independent or external validation. Sensitivity results, if run, use the

taxonomy-recognized subset (6,785 rows) with grouped five-fold CV; they overlap the source data and are not fully nested post-selection validation.

Importance, local explanation, and application boundary

Global importance is Random Forest MDI. It is non-negative split-usage importance, not a signed effect, causal estimate, ingredient toxicity score, or safety certification. Local interpretation measures the probability change after dropping one supplied ingredient at a time; it is **not SHAP**. The Streamlit interface is local only. No hosted service, OCR flow, full synonym resolution, deterministic registry score, or end-to-end regulatory assessment is claimed.

Reproducibility and evidence

- Final dynamic metrics
- Final readable metrics summary
- Saved test evidence: 23 passed
- Notebook execution evidence: 43 notebook cells, 24 executed code cells, 0 error outputs, 15 embedded PNG figures.
- Random Forest notebook

A future claim of independent label recreation requires the complete versioned registry and matching procedure plus independent labels.